

Tanvi Garg

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EDUCATION

Virginia Tech

M.S. Thesis in Computer Engineering | GPA: 4.0 / 4.0

Honors: CCI Southwest Virginia (SWVA) Scholarship Recipient – 2025, 2026

Arlington, VA

Aug 2023 – Dec 2026

Virginia Tech

M.Eng. in Computer Science | GPA: 3.80 / 4.0

Blacksburg, VA

Aug 2021 – May 2023

NMIMS

B.Tech. in Computer Science and Engineering | GPA: 3.45 / 4.0

Mumbai, India

Jul 2017 – May 2021

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

AI/ML & Security: Docker, Kubernetes, AWS, Helm, RKE, Multus CNI, Git, Databricks

Cloud & Infrastructure: LINDDUN, STRIDE, threat modeling, HIPAA compliance, role-based access control, mCODE FHIR R4, OWASP, Nmap, Wireshark

Networking & Systems: TCP/IP, OpenAirInterface, srsRAN, FlexRIC, ZeroMQ, Wireshark, USRP X310

EXPERIENCE

Graduate Research Assistant

Aug 2023 – Present

Networks and Systems Security Lab, Virginia Tech | Advisor: Dr. Angelos Stavrou

Arlington, VA

- Developed security and privacy mechanisms for distributed AI systems, including middleware for enforcing role-scoped permissions across inter-agent communication boundaries.
- Conducted root-cause and statistical analysis of LLM-based prompt-injection detection failures, identifying **positional brittleness as a previously undocumented failure mode**.
- Analyzed real-time telemetry from an **OpenAirInterface 5G NR testbed with USRP X310** for anomaly detection and closed-loop RAN performance optimization.

Research Programmer

Jun 2022 – Aug 2023

Virginia Tech

Arlington, VA

- Dockerized** Wave, a decentralized authorization framework with transitive delegation, enabling reproducible self-service research environments.
- Deployed a distributed **5G Core and RAN on Kubernetes** using Docker Compose, Helm, Multus CNI, and RKE to evaluate network slicing and UE performance.
- Built **Python and SQL data pipelines** to ingest real-time system telemetry and identify root causes of performance degradation.

SELECTED PROJECTS

RoleGuard: Privacy Enforcement for Multi-Agent LLM Systems

2026

- Formalized **role-boundary leakage**, a privacy failure in multi-agent LLM pipelines where agents receive sensitive information beyond their permitted scope.
- Architected **RoleGuard**, an inference-time middleware layer enforcing role-scoped access control at every inter-agent boundary without retraining or modifying underlying LLM agents.
- Built **RoleLeak**, a two-tier evaluation benchmark spanning **180 mCODE synthetic patients**, reducing PHI boundary violations from **100% to 0% while preserving 100% task completion**.
- Designed a cyclic **LangGraph multi-agent pipeline** and evaluated adversarial robustness against prompt injection, role impersonation, and targeted PHI-fishing attacks; **RoleGuard blocked all evaluated attacks**.

AI-Powered Weight Management Platform

In Progress, 2026

- Developing a full-stack application to track nutrition, exercise, activity, hydration, sleep, and body measurements and provide AI-assisted weekly progress recommendations.
- Building toward a **Python REST API, relational data layer, React/TypeScript interface, and containerized deployment**; implementation in progress.

O-RAN xApp Development

2023

- Deployed a **Near-RT RIC environment** using srsRAN, FlexRIC, and ZeroMQ; implemented xApps for real-time signal metrics and used Wireshark to isolate a persistent UE-gNB attachment failure to the ZeroMQ RF driver.